

Introduction

Recall $SE(\hat{\beta}_j) = \sqrt{\text{var}(\hat{\beta}_j)}$ from last lecture

Assuming

$$u_i \sim N(0, \sigma^2)$$

Classical error assumption: independent normality

Confidence intervals and hypothesis testing

Normal distribution

Because σ^2 unknown, use t distribution

Same idea with confidence intervals

Hypothesis testing

Takeaways

Key takeaways

Appendix

During $\beta_2 \sim$

$$N\left(\beta_2, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$$

χ^2 distribution

t distribution

ECONOMETRICS I

Lecture 5

Understanding statistical significance

Matías Cabello

matias.cabello@wiwi.uni-halle.de

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Slides

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Hypothesis testing

Takeaways

Key takeaways

Appendix

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 χ^2 distribution
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Tech Adoption: Linear vs Log-Linear Models

	Dependent variable:	
	Users Linear (1)	log(Users) Log-Linear (2)
Year	200.094*** (25.205)	0.241*** (0.022) ← $\hat{\beta}_2$ estimate
Constant	-401,968.400*** (50,787.630)	-479.229*** (44.104) ← $\hat{\beta}_1$ estimate
Observations	21	21
R2	0.768	0.864 ← R^2

Note: *p<0.1; **p<0.05; ***p<0.01

Introduction

Slides

Introduction

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Classical error assumption: independent normality

Confidence intervals and hypothesis testing

Normal distribution

Because σ^2 unknown, use t distribution

Same idea with confidence intervals

Hypothesis testing

Takeaways

Key takeaways

Appendix

During $\hat{\beta}_j \sim$

$N\left(\beta_j, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$

χ^2 distribution

t distribution

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Standard errors

Introduction

Slides

Introduction

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Assuming $u_i \sim N(0, \sigma^2)$

Classical error assumption: independent normality

Confidence intervals and hypothesis testing

Normal distribution

Because σ^2 unknown, use t distribution

Same idea with confidence intervals

Hypothesis testing

Takeaways

Key takeaways

Appendix

During $\beta_j \sim$

$N\left(\beta_j, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$
 χ^2 distribution
 t distribution

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What is this?

Introduction

Slides

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Assuming

$u_i \sim N(0, \sigma^2)$

Classical error assumption: independent normality

Confidence intervals and hypothesis testing

Normal distribution

Because σ^2 unknown, use t distribution

Same idea with confidence intervals

Hypothesis testing

Takeaways

Key takeaways

Appendix

Distributing $\hat{\beta}_j$ ~

$N\left(\beta_j, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$

χ^2 distribution

t distribution

The stars denote **statistical significance** at different levels (1%, 5%, 10%).

Similarly to $s(\hat{\beta}_j) = \sqrt{\text{var}(\hat{\beta}_j)}$, they represent a measure of precision (but focusing on how certain we are that $\hat{\beta}_j \neq 0$).

Today's objective: understand what these stars mean and the concept of statistical significance and confidence intervals more generally.

Recall $s(\hat{\beta}_j) = \sqrt{\text{var}(\hat{\beta}_j)}$ from last lecture

Slides

Introduction

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Assuming

$u_i \sim N(0, \sigma^2)$

Classical error assumption: independent normality

Confidence intervals and hypothesis testing

Normal distribution

Because σ^2 unknown, use t -distribution

Same idea with confidence intervals

Hypothesis testing

Takeaways

Key takeaways

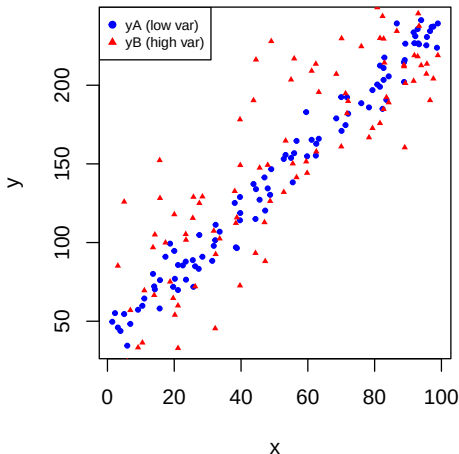
Appendix

Deriving $\hat{\beta}_2 \sim$

$N\left(\beta_2, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$

χ^2 -distribution

t -distribution



Different error terms:

- Low-variance u_A vs. high-variance u_B .
- Both normally distributed

$$u_A \sim N(0, 10^2)$$

$$u_B \sim N(0, 40^2)$$

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Slides

Introduction

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Assuming $u_i \sim N(0, \sigma^2)$

Classical error assumption: independent normality

Confidence intervals and hypothesis testing

Normal distribution

Because σ^2 unknown, use t distribution

Same idea with confidence intervals

Hypothesis testing

Takeaways

Key takeaways

Appendix

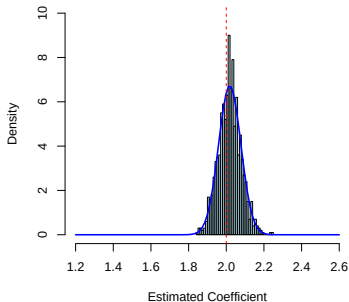
During $\hat{\beta}_2 \sim$

$N(\beta_2, \frac{\sigma^2}{\sum (x_i - \bar{x})^2})$

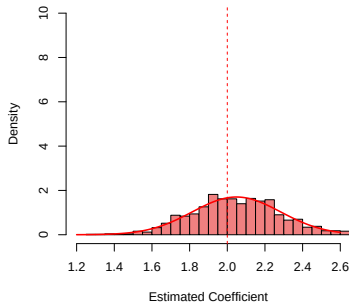
χ^2 distribution

t distribution

Sampling Distribution of β (yA)



Sampling Distribution of β (yB)



Note how error variance σ^2 is “inherited” by the $\text{var}(\hat{\beta}_2)$

Recall $s(\hat{\beta}_j) = \sqrt{\text{var}(\hat{\beta}_j)}$ from last lecture

$$\text{var}(\hat{\beta}_2|x) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}$$

or

$$\text{var}(\hat{\beta}_2|x) = \frac{\sigma^2}{n \text{ var}(x)}$$

We saw that **precision** will be higher (the variance lower):

- with more data ($\uparrow n$)
- with lower error variance ($\downarrow \sigma^2$)

Slides

Introduction

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Assuming

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Classical error assumption: independent normality

Confidence intervals and hypothesis testing

Normal distribution

Because σ^2 unknown, use t distribution

Same idea with confidence intervals

Hypothesis testing

Takeaways

Key takeaways

Appendix

During $\hat{\beta}_2 \sim$

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χ^2 distribution
 t distribution

$$\text{ASSUMING } u_i \sim N(0, \sigma^2)$$

Classical error assumption: independent normality

Data generating process (population) with

$$y_i = \beta_1 + \beta_2 x_i + u_i$$

and **classical error specification**:

$$u_i|x \sim N(0, \sigma^2) \quad \text{for } i = 1, \dots, n$$

Shorthand for:

- **normality**: $u_i|x \sim \text{Normal}(\mathbb{E}[u_i|x], \text{var}[u_i|x])$
- **zero mean**: $\mathbb{E}[u_i|x] = 0$ for all i
- **homoskedasticity**: $\text{var}(u_i|x) = \sigma^2$ for all i
- **uncorrelatedness**: $\text{cov}(u_i, u_j|x) = 0$ for all $i \neq j$

all **conditional on $\mathbf{x}$** ($\dots|x$) (like in an experimental setting).

Classical error assumption: independent normality

Slides

Introduction

Recall $S(\hat{\beta}_j) = \sqrt{\text{var}(\hat{\beta}_j)}$ from last lecture

Assuming $u_i \sim N(0, \sigma^2)$

Classical error assumption: independent normality

Confidence intervals and hypothesis testing

Normal distribution

Because σ^2 unknown, use t distribution

Same idea with confidence intervals

Hypothesis testing

Takeaways

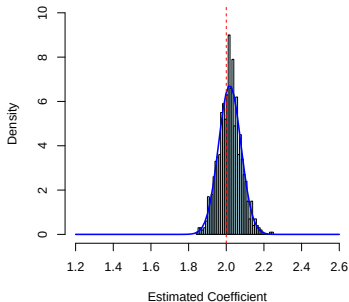
Key takeaways

Appendix

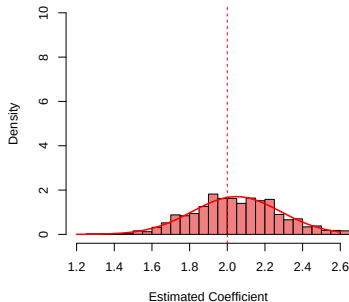
Deriving $\hat{\beta}_2 \sim$

$N\left(\beta_2, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$
 χ^2 distribution
 t distribution

Sampling Distribution of β (y_A)



Sampling Distribution of β (y_B)



Plots suggest: **normality** of u_i is “inherited” by estimator $\hat{\beta}_2$

Classical error assumption: independent normality

Indeed, we can show (proof in the appendix):

$$\hat{\beta}_2|x \sim N\left(\beta_2, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$$

- In other words: **normal errors** $\implies$ **normal** $\hat{\beta}_2$

- Note two familiar results:

$$\mathbb{E}(\hat{\beta}_2|x) = \beta_2 \quad (\text{unbiasedness})$$

$$\text{var}(\hat{\beta}_2|x) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2} \quad (\text{as before})$$

- Since $\hat{\beta} \sim \text{normal}$, **we can compute probabilities based on $N(0, 1)$**

Introduction

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Assuming

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Classical error assumption: independent normality

Confidence intervals and hypothesis testing

Normal distribution

Because σ^2 unknown, use t -distribution

Same idea with confidence intervals

Hypothesis testing

Takeaways

Key takeaways

Appendix

During $\beta_2 \sim$

$$N\left(\beta_2, \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}\right)$$

χ^2 -distribution
 t -distribution

CONFIDENCE INTERVALS AND HYPOTHESIS TESTING

Normal distribution

Slides

Introduction

Recall $S(\beta_j) = \sqrt{\text{var}(\beta_j)}$ from last lecture

Assuming $u_i \sim N(0, \sigma^2)$

Classical error assumption: independent normality

Confidence intervals and hypothesis testing

Normal distribution

Because σ^2 unknown, use t -distribution

Same idea with confidence intervals

Hypothesis testing

Takeaways

Key takeaways

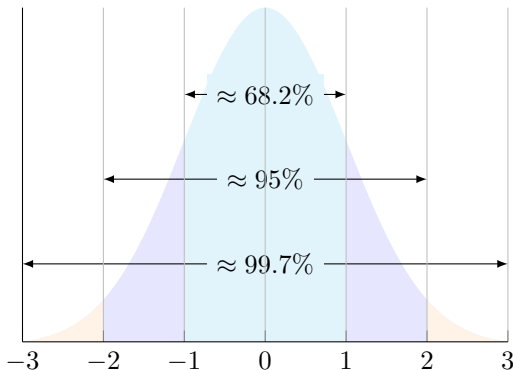
Appendix

Douglas $\beta_2 \sim$

$N\left(\beta_2, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$

χ^2 distribution

t distribution



Standard deviations away from the mean

In a normally distributed variable:

- $\approx 95\%$ of observations fall within **2 st. dev.**
- over 99% of observations fall within **3 st. dev.**

Normal distribution

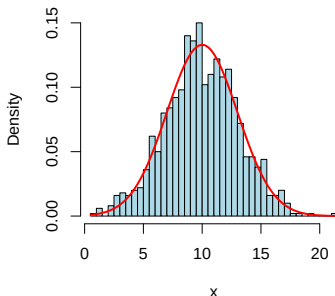
You can always rescale a normally distributed variable

$$x \sim N(\mu, \sigma^2)$$

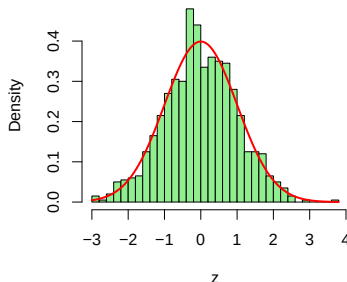
into a standard normal variable

$$z = \frac{x - \mu}{\sigma} \sim N(0, 1)$$

Original: $N(10, 9)$



Standardized: $N(0, 1)$



Normal distribution

Slides

Introduction

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Assuming

$u_i \sim N(0, \sigma^2)$

Classical error assumption:
independent normality

Confidence intervals
and hypothesis testing

Normal distribution

Because σ^2 unknown, use
 t distribution

Same idea with confidence
intervals

Hypothesis testing

Takeaways

Key Takeaways

Appendix

Deriving $\beta_2 \sim$

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χ^2 distribution

t distribution

R code generating the previous graphs:

```
# Generate random normal data
set.seed(1) # for reproducibility
n <- 1000
x <- rnorm(n, mean = 10, sd = 3) # N(10, 9)

# Standardize
z <- (x - mean(x)) / sd(x)

# Create plots
par(mfrow = c(1, 2))

# Original data with N(10, 9) density
hist(x, breaks = 30, probability = TRUE, main = "Original: -N(10,-9)", xlab =
  "x", col = "lightblue")
curve(dnorm(x, mean = 10, sd = 3), add = TRUE, col = "red", lwd = 2)

# Standardized data with N(0, 1) density
hist(z, breaks = 30, probability = TRUE, main = "Standardized: -N(0,-1)",
  xlab = "z", col = "lightgreen")
curve(dnorm(x, mean = 0, sd = 1), add = TRUE, col = "red", lwd = 2)
```


Normal distribution

[Slides]

Introduction

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Classical error assumption: independent normality

Confidence intervals and hypothesis testing

Normal distribution

Because σ^2 unknown, use t -distribution

Same idea with confidence intervals

Hypothesis testing

Takeaways

Key takeaways

Appendix

During $\beta_2 \sim$

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 χ^2 -distribution
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Why know that:

$$\hat{\beta}_2 \sim N\left(\beta_2, \frac{\sigma^2}{\sum(x_i - \bar{x})^2}\right)$$

Standardizing gives:

$$\frac{\hat{\beta}_2 - \beta_2}{\sqrt{\frac{\sigma^2}{\sum(x_i - \bar{x})^2}}} \sim N(0, 1)$$

Normal distribution

Slides

Introduction

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Classical error assumption: independent normality

Confidence intervals and hypothesis testing

Normal distribution

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Same idea with confidence intervals

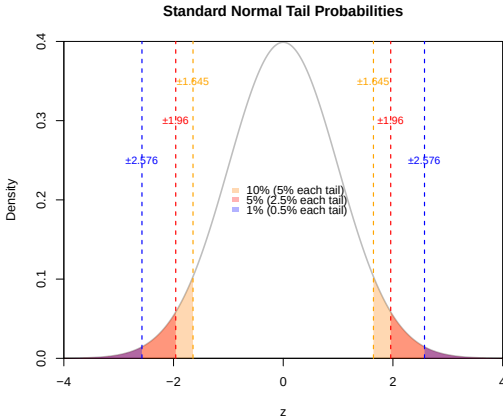
Hypothesis testing

Takeaways

Key takeaways

Appendix

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 χ^2 distribution
 t distribution



Confidence intervals

- $z \approx 2.576$
99% C.I.
0.5% each tail
- $z \approx 1.96$
95% C.I.
2.5% each tail
- $z \approx 1.645$
90% C.I.
5% each tail

Normal distribution

Inequality vs. income

	Dependent variable:	
	income2010 (1)	(2)
income1950	1.792** (0.704)	
ineq1950		-0.019*** (0.004)
Constant	0.012** (0.004)	0.060*** (0.009)
Observations	13	13
R2	0.371	0.664

Note: *p<0.1; **p<0.05; ***p<0.01

Yet this is just an approximation!

Using a normal approximation:

At which level of significance could we reject the hypothesis that $\beta_{\text{income1950}} = 0$ after obtaining $\hat{\beta}_{\text{income1950}} = 1.792$?

$$1.792/.704 = 2.54 > 1.96 \text{ but } < 2.576 \implies 5\%$$

At which level of significance could we reject the hypothesis that $\beta_{\text{ineq1950}} = 0$ after obtaining $\hat{\beta}_{\text{ineq1950}} = -0.019$?

$$.019/.004 = 4.75 > 2.576 \implies 1\%$$

Because σ unknown: use t distribution

When σ^2 is **known**, under classical assumptions:

$$\hat{\beta}_2 \sim N\left(\beta_2, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$$

Standardizing gives:

$$\frac{\hat{\beta}_2 - \beta_2}{\sqrt{\frac{\sigma^2}{\sum (x_i - \bar{x})^2}}} \sim N(0, 1)$$

But in practice, σ^2 is **unknown** and must be estimated.

Problem: We need to replace σ^2 with an estimator $\hat{\sigma}^2$.

Because σ unknown: use t distribution

We estimate σ^2 using:

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2$$

Key results:

- $\hat{\sigma}^2$ is unbiased: $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$
- $\frac{(n-2)\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-2}^2$ (details in appendix)
- $\hat{\beta}_2$ and $\hat{\sigma}^2$ are independent

Therefore (details in appendix):

$$\frac{\hat{\beta}_2 - \beta_2}{\sqrt{\frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2}}} = \frac{\frac{\hat{\beta}_2 - \beta_2}{\sigma^2}}{\sqrt{\frac{\hat{\sigma}^2}{\sigma^2 \sum (x_i - \bar{x})^2}}} \sim \frac{N(0, 1)}{\sqrt{\frac{\chi_{n-2}^2}{n-2}}} \sim t_{n-2}$$

Because σ unknown: use t distribution

Slides

Introduction

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Assuming $u_i \sim N(0, \sigma^2)$

Classical error assumption: independent normality

Confidence intervals and hypothesis testing

Normal distribution

Because σ unknown: use t distribution

Same idea with confidence intervals

Hypothesis testing

Takeaways

Key takeaways

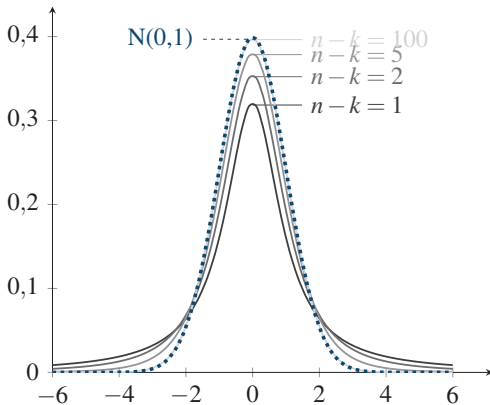
Appendix

Deriving $\beta_2 \sim$

$N\left(\beta_2, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$

χ^2 distribution

t distribution

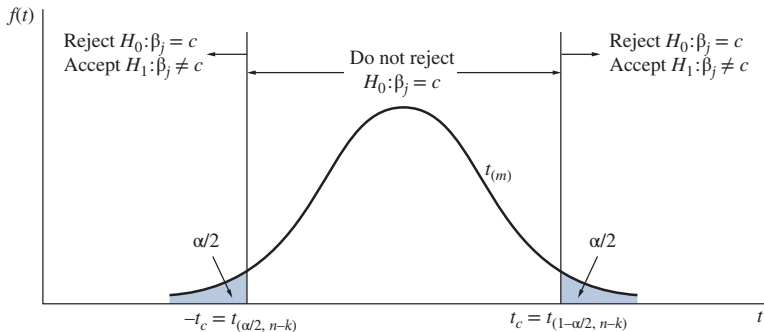


The t -distribution:

- Becomes standard normal $N(0,1)$ for large $n-k$.
- Thus: under high n , you can approximate as if normal.

Because σ unknown: use t distribution

Rejection zone for $H_0 : \beta_j = c$ at significance level α



The test statistic is $\hat{t} = \frac{\hat{\beta}_j - \beta_j^{H_0}}{\hat{s}(\hat{\beta}_j)}$

Because σ unknown: use t distribution

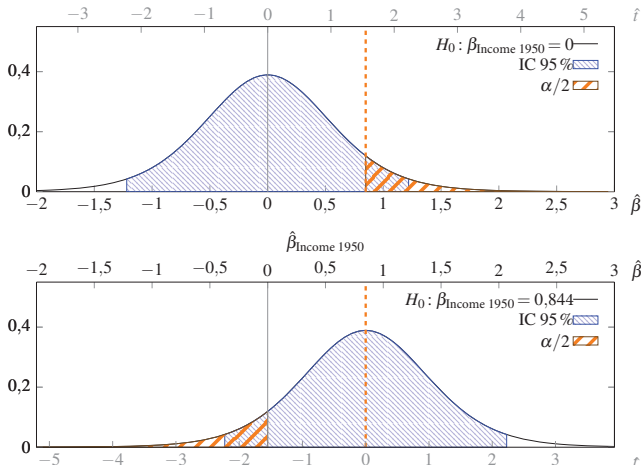
The stars in R correspond to the case $\beta_j^{H_0} = 0$ at $\alpha = 10\%, 5\%, 1\%$.

Inequality vs. income

	Dependent variable:					
	income2010					
	(1)	(2)	(3)	(4)	(5)	(6)
income1950	1.792** (0.704)		2.454 (1.639)		0.256 (1.354)	0.844 (0.552)
ineq1950		-0.019*** (0.004)		-0.018** (0.005)	-0.017* (0.007)	-0.016*** (0.004)
Constant	0.012** (0.004)	0.060*** (0.009)	0.007 (0.010)	0.057*** (0.009)	0.054* (0.020)	0.049*** (0.011)
Observations	13	13	6	6	6	13
R2	0.371	0.664	0.359	0.799	0.801	0.728

Note: *p<0.1; **p<0.05; ***p<0.01

Same idea with confidence intervals



$$95\% \text{ CI: } \left(\hat{\beta}_j \pm t_{n-k, 2.5\%} \cdot \sqrt{\widehat{\text{var}}[\hat{\beta}_j]} \right)$$

Hypothesis testing

Statistical significance is used also more generally in the context of hypothesis testing.

Components of Hypothesis Tests

- 1 A null hypothesis H_0 (e.g., $H_0 : \beta_2 = 0$)
- 2 An alternative hypothesis H_1 (e.g., $H_1 : \beta_2 \neq 0$)
- 3 A test statistic (e.g., $\hat{t} = \frac{\hat{\beta}_2}{\hat{s}(\hat{\beta}_2)}$)
- 4 A rejection region (e.g., $|\hat{t}| > t_{n-k, 2.5\%} = 2.1$)
- 5 A conclusion (e.g., reject H_0 at $\alpha = 5\%$ because $|\hat{t}| > 2.1$)

Introduction

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$$u_i \sim N(0, \sigma^2)$$

Classical error assumption:
independent normality

Confidence intervals
and hypothesis testing

Normal distribution

Because σ^2 unknown, use
 t distribution

Same idea with confidence
intervals

Hypothesis testing

Takeaways

Key takeaways

Appendix

During $\hat{\beta}_2 \sim$

$$N\left(\beta_2, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$$

χ^2 distribution
 t distribution

TAKEAWAYS

Key takeaways

You should now know:

- Transform normal to standard normal
- Key reference values: e.g. 95% CI within 2 st. deviations
- That $u|x \sim N(0, \sigma^2)$ implies conditional homoskedasticity, zero mean, and no autocorrelation.
- That $u|x \sim N(0, \sigma^2)$ leads to $\hat{\beta}_2|x \sim N(\beta, \sigma^2 / \sum(x_i - \bar{x})^2)$
- That $\frac{\hat{\beta}_2 - \beta_2}{s(\hat{\beta}_2)} \sim N(0, 1)$ but $\frac{\hat{\beta}_2 - \beta_2}{\hat{s}(\hat{\beta}_2)} \sim t_{n-2}$
- That stars in R denote α for rejecting null hypothesis $H_0 : \hat{\beta}_j = 0$ based on t statistic

Introduction

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Classical error assumption: independent normality

Confidence intervals and hypothesis testing

Normal distribution

Because σ^2 unknown, use t distribution

Same idea with confidence intervals

Hypothesis testing

Takeaways

Key takeaways

Appendix

During $\hat{\beta}_2 \sim$

$$N\left(\beta_2, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$$

χ^2 distribution
 t distribution

APPENDIX

Deriving $\hat{\beta}_2 \sim N\left(\beta_2, \frac{\sigma^2}{\sum(x_i - \bar{x})^2}\right)$

Normal Distribution Transformation Rule

For any scalar random variable $Y \sim N(\mu, \sigma^2)$:

$$aY + b \sim N(a\mu + b, a^2\sigma^2)$$

Application to Regression:

- Errors: $u_i \sim N(0, \sigma^2)$
- Then $y_i = \beta_1 + \beta_2 x_i + u_i \sim N(\beta_1 + \beta_2 x_i, \sigma^2)$
- ... and

$$\hat{\beta}_2 = \frac{\text{cov}(x, y)}{\text{var}(x)} \sim N\left(\beta_2, \frac{\sigma^2}{\sum(x_i - \bar{x})^2}\right)$$

(proof follows)

Deriving $\hat{\beta}_2 \sim N\left(\beta_2, \frac{\sigma^2}{\sum(x_i - \bar{x})^2}\right)$

How do we obtain $\hat{\beta}_2 \sim N\left(\beta_2, \frac{\sigma^2}{\sum(x_i - \bar{x})^2}\right)$?

1. Rewrite OLS formula:

$$\begin{aligned}\hat{\beta}_2 &= \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} \\ &= \frac{\sum(x_i - \bar{x})y_i - \overbrace{\sum(x_i - \bar{x})\bar{y}}^{0, \text{ since } \sum(x_i - \bar{x})=0}}{\sum(x_i - \bar{x})^2} = \sum w_i y_i\end{aligned}$$

where $w_i = \frac{(x_i - \bar{x})}{\sum(x_i - \bar{x})^2}$. Thus $\hat{\beta}_2$ = linear combination of y_i .

2. Since y_i are normal, $\hat{\beta}_2$ is a linear combination of normals:

$$\hat{\beta}_2 \sim N\left(\sum w_i E[y_i], \sum w_i^2 \text{var}(y_i)\right)$$

Deriving $\hat{\beta}_2 \sim N\left(\beta_2, \frac{\sigma^2}{\sum(x_i - \bar{x})^2}\right)$

3. Determine mean and variance of that $N(\cdot)$

Claim 1: The mean $\sum w_i E[y_i]$ is equal to β_2 .

Proof: This follows directly from the unbiasedness of $\hat{\beta}_2$.

Claim 2: The variance $\sum w_i^2 \text{var}(y_i)$ is equal to $\frac{\sigma^2}{\sum(x_i - \bar{x})^2}$.

Proof: We already know that $\text{var}(y_i) = \sigma^2$. What about $\sum w_i^2$?

$$\sum w_i^2 = \sum \frac{(x_i - \bar{x})^2}{(\sum(x_i - \bar{x})^2)^2} = \frac{1}{\sum(x_i - \bar{x})^2}$$

Thus:

$$\hat{\beta}_2|x \sim N\left(\beta_2, \frac{\sigma^2}{\sum(x_i - \bar{x})^2}\right)$$

χ^2 distribution

[Slides]

Introduction

Recall $S(\beta_j) = \sqrt{\text{var}(\beta_j)}$ from last lecture

Assuming

$u_i \sim N(0, \sigma^2)$

Classical error assumption: independent normality

Confidence intervals and hypothesis testing

Normal distribution

Because σ^2 unknown, use t -distribution

Same idea with confidence intervals

Hypothesis testing

Takeaways

Key takeaways

Appendix

Distributing $\beta_j \sim$

$N\left(\beta_j, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$

χ^2 distribution

t -distribution

- **χ^2 Distribution:** If $Z_1, Z_2, \dots, Z_n$ are independent standard normal variables, then

$$Z = \sum_{i=1}^n Z_i^2 \sim \chi_n^2 \quad \text{and} \quad \sqrt{Z} \sim \chi_n.$$

(Note: if k of the n normal variables are not independent, then $\sim \chi_{n-k}^2$)

t distribution

Slides

Introduction

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Appendix

During $\beta_j \sim$

$N\left(\beta_j, \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}\right)$

χ^2 distribution

t distribution

- **t Distribution:** Let Z_1 be a standard normal variable and Z_2 be a χ^2 variable with n degrees of freedom, that is $Z_1 \sim N(0, 1)$ and $Z_2 \sim \chi_n^2$, then

$$t = \frac{Z_1}{\sqrt{Z_2/n}} = \frac{\sqrt{n}Z_1}{\sqrt{Z_2}} \sim t_n.$$

- Thus, in our regression

$$\frac{\hat{\beta}_j - \beta_{H0}}{\hat{s}(\beta_j)} \sim t_{n-k}$$